Linear Models II

Linear Algebra · Lecture 12

Last time (Lecture 11): $\beta^0 = GX'y$ (G a g-inverse of $X'X$) solves the normal equations; $H = XGX'$ does not depend on G; $\hat{\sigma}^2 = y'My/(n-r)$ ($M = I_n - H$) is unbiased. Earlier results are cited where they are used; F1–F3: probability, no proof. e is the vector of residuals; unit vectors are columns of I.

F1. $\mathsf{P}(u \notin \mathcal{A}) = 1 - \mathsf{P}(u \in \mathcal{A})$; $u_1, \dots, u_m$ are **independent** if $\mathsf{P}(u_k \in \mathcal{A}_k \ \forall k) = \prod_k \mathsf{P}(u_k \in \mathcal{A}_k)$; then so are $g_1(u_1), \dots, g_m(u_m)$.

F2. χ^2_r, t_ν, $F_{a,b}$ (**degrees of freedom** $r, \nu, a, b \geqslant 1$): the distributions of $\sum_{i \leqslant r} w_i^2$, $Z/\sqrt{Y/\nu}$, $(Y_1/a)/(Y_2/b)$ for independent $w_i, Z \sim N(0,1)$, $Y \sim \chi^2_\nu$, $Y_1 \sim \chi^2_a$, $Y_2 \sim \chi^2_b$.

F3. For u as in F2: $\mathsf{P}(u = c) = 0$; the **quantile** $t_{\nu,\tau}$, $F_{a,b,\tau}$ ($0 < \tau < 1$) is the unique c with $\mathsf{P}(u \leqslant c) = \tau$.

1. Estimable functions and their best linear unbiased estimators

Definition 1. In the linear model $y = X\beta + \varepsilon$ (Lecture 11, Definition 1: $X \in \mathbb{R}^{n \times p}$ known, $\beta \in \mathbb{R}^p$ unknown, $\varepsilon \in \mathbb{R}^n$ random), with X of any rank, assume (A1) $\mathsf{E}(y) = X\beta$, and let $a \in \mathbb{R}^p$. A **linear estimator** is a random variable $c'y$ with a constant $c \in \mathbb{R}^n$; it is a **linear unbiased estimator** of $a'\beta$ if $\mathsf{E}(c'y) = a'\beta$ for every $\beta \in \mathbb{R}^p$. $a'\beta$ is **estimable** if it has a linear unbiased estimator. A linear unbiased estimator of $a'\beta$ with the smallest variance among all of them is a **best linear unbiased estimator** (**BLUE**) of $a'\beta$.

Theorem 2. Assume (A1), and let $a \in \mathbb{R}^p$.

(a) A linear estimator $c'y$ is unbiased for $a'\beta \iff X'c = a$. Hence $a'\beta$ is estimable $\iff a \in \mathcal{C}(X') \iff a \perp \mathcal{N}(X)$. If $\operatorname{rank} X = p$, every $a'\beta$ is estimable.

(b) Let G be a g-inverse of $X'X$ (Lecture 11, Definition 8: $AGA = A$ for $A = X'X$) and $H = XGX'$. If $a = X'c$, then $a'\hat{\beta} = c'\hat{y} = c'Hy$ ($\hat{y} = X\hat{\beta}$) for every least-squares estimate $\hat{\beta}$, a minimizer of $\|y - X\beta\|^2$ (Lecture 11, Definition 1): the same linear estimator for all of them; and $\mathsf{E}(a'\hat{\beta}) = a'\beta$.

(c) If $a'\beta$ is not estimable, then $a'\hat{\beta}$ takes every real value as $\hat{\beta}$ runs through the least-squares estimates.

Proof.

(a) $\mathsf{E}(c'y) = c'\,\mathsf{E}(y) = c'X\beta$ (Lecture 9, Theorem 3(b): $\mathsf{E}(Az + b) = A\,\mathsf{E}(z) + b$; (A1)); $z \perp W$ means $z'w = 0$ for all $w \in W$ (Lecture 6, Definition 4); $\operatorname{rank} A = \dim \mathcal{C}(A)$ (Lecture 4, Definition 10); subspaces $U \subseteq W$ with $\dim U = \dim W$ are equal (Lecture 4, Theorem 5(a)):

$$
\begin{aligned}
c'X\beta = a'\beta \text{ (all } \beta) &\implies c'X \operatorname{col}_j(I_p) = a' \operatorname{col}_j(I_p) \text{ (all } j) && \text{(Lecture 1, Proposition 11(a))}\\
&\implies c'X = a' \iff X'c = a && \text{(Lecture 1, Theorem 8)}\\
X'c = a &\implies \mathsf{E}(c'y) = c'X\beta = a'\beta &&\\
&\implies \big(a'\beta \text{ estimable} \iff a \in \{X'c\} = \mathcal{C}(X')\big) && \text{(Lecture 4, Definition 7)}\\
\mathcal{N}(X) = \mathcal{C}(X')^\perp &\implies \mathcal{C}(X') \subseteq \mathcal{N}(X)^\perp && \text{(Lecture 6, Corollary 6)}\\
\dim \mathcal{C}(X') = \operatorname{rank} X' = \operatorname{rank} X &= p - \dim \mathcal{N}(X) && \text{(Lecture 4, Theorems 12(a), 11)}\\
&= \dim \mathcal{N}(X)^\perp \implies \mathcal{C}(X') = \mathcal{N}(X)^\perp && \text{(Lecture 6, Theorem 5(c))}\\
\operatorname{rank} X = p &\implies \operatorname{rank} X' = p \implies \mathcal{C}(X') = \mathbb{R}^p && \text{(Lecture 4, Theorem 12(a),(c))}
\end{aligned}
$$

(b) $a = X'c$; $\mathsf{E}(c'Hy) = c'H\,\mathsf{E}(y)$ (Lecture 9, Theorem 3(b)) and (A1); all least-squares estimates give the same $\hat{y} = X\hat{\beta}$ (Lecture 11, Theorem 3(b)):

$$X\hat{\beta} = \hat{y} = Hy \implies a'\hat{\beta} = c'X\hat{\beta} = c'\hat{y} = c'Hy \qquad \text{(Lecture 11, Theorem 10(e))}$$
$$\mathsf{E}(c'Hy) = c'HX\beta = c'X\beta = a'\beta \qquad \text{(Lecture 11, Theorem 10(c))}$$

(c) By (a), $a \notin \mathcal{N}(X)^{\perp}$; $X'X\beta = X'y$ has a solution $\hat{\beta}$ (Lecture 11, Theorem 2(a)); the least-squares estimates are the solutions of $X'X\beta = X'y$ (Lecture 11, Theorem 3(a)):

$a'z \neq 0$ for some $z \in \mathcal{N}(X)$

$Xz = 0 \implies X'X(\hat{\beta} + tz) = X'X\hat{\beta} = X'y \quad (t \in \mathbb{R}) \implies \hat{\beta} + tz$ is a least-squares estimate

$a'(\hat{\beta} + tz) = a'\hat{\beta} + t\,a'z$ takes every real value

□

Example 3. In Lecture 11, Example 7, the one-way model $y_i = \mu + \alpha_l + \varepsilon_i$ of Lecture 11, Theorem 6 has $\beta = (\mu, \alpha_1, \alpha_2, \alpha_3)'$, $y = (5, 7, 6, 3, 5, 2)'$ in groups of sizes $(n_1, n_2, n_3) = (3, 2, 1)$, $X'X = \begin{pmatrix} 6&3&2&1\\ 3&3&0&0\\ 2&0&2&0\\ 1&0&0&1 \end{pmatrix}$, $X'y = (28, 18, 8, 2)'$ and the least-squares estimates $(-t, 6+t, 4+t, 2+t)'$ $(t \in \mathbb{R})$. Which of $\mu + \alpha_1$, $\alpha_1 - \alpha_2$, μ, α_1 are estimable? With the group means $\bar{y}_1 = \frac{1}{3}(y_1 + y_2 + y_3)$, $\bar{y}_2 = \frac{1}{2}(y_4 + y_5)$, $\bar{y}_3 = y_6$, find the estimator of $\alpha_1 - \alpha_2$ of Theorem 2(b).

Solution.

$\mathcal{N}(X) = \operatorname{span}\{(-1, 1, 1, 1)'\}$ (Lecture 11, Theorem 6(b)); Theorem 2(a),(c):

$a_0\mu + a_1\alpha_1 + a_2\alpha_2 + a_3\alpha_3$ is estimable $\iff -a_0 + a_1 + a_2 + a_3 = 0$

$\mu + \alpha_1 :\ -1 + 1 = 0, \quad \alpha_1 - \alpha_2 :\ 1 - 1 = 0 \implies$ estimable; on $(-t, 6+t, 4+t, 2+t)'$: values $6,\ 2$

$\mu :\ -1 \neq 0, \quad \alpha_1 :\ 1 \neq 0 \implies$ not estimable; values $-t,\ 6+t$ change with t

$X'y$ holds the total and the group totals $\operatorname{col}_j(X)'y$ (Lecture 1, Theorem 4(d), Definition 7); Lecture 11, Theorem 3(a):

$$X'X(0, \bar{y}_1, \bar{y}_2, \bar{y}_3)' = (3\bar{y}_1 + 2\bar{y}_2 + \bar{y}_3,\ 3\bar{y}_1,\ 2\bar{y}_2,\ \bar{y}_3)' = X'y$$
$$\hat{\beta} = (0, \bar{y}_1, \bar{y}_2, \bar{y}_3)' \text{ is a least-squares estimate}$$
$$\hat{\alpha}_1 - \hat{\alpha}_2 = \bar{y}_1 - \bar{y}_2 = c'y, \qquad c = \left(\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}, -\tfrac{1}{2}, -\tfrac{1}{2}, 0\right)'$$
$$X'c = (1 - 1,\ 1,\ -1,\ 0)' = (0, 1, -1, 0)'$$

Theorem 4. Assume (A1)–(A3) (Lecture 11, Definition 1: (A2) $\mathsf{E}(\varepsilon) = 0$, (A3) $\operatorname{var}(\varepsilon) = \sigma^2 I_n$, $\sigma^2 > 0$), with X of any rank; let G be a g-inverse of $X'X$, $H = XGX'$, $M = I_n - H$, and let $a'\beta$ be estimable. Then $a'\hat{\beta}$, the same for every least-squares estimate $\hat{\beta}$, is the BLUE of $a'\beta$: it is a linear unbiased estimator of $a'\beta$, and every linear unbiased estimator $c'y$ of $a'\beta$ has $\operatorname{var}(c'y) \geqslant \operatorname{var}(a'\hat{\beta})$, with equality only if $c'y = a'\hat{\beta}$ for every y. Moreover $\operatorname{var}(a'\hat{\beta}) = \sigma^2 a'Ga$ (for every such G).

Proof.

$c'y$ a linear unbiased estimator of $a'\beta$; (A3) gives $\operatorname{var}(y) = \sigma^2 I_n$ (Lecture 11, Definition 1); $\operatorname{var}(Az+b) = A\operatorname{var}(z)A'$ (Lecture 9, Theorem 3(b)); $H' = H = H^2$, $M' = M = M^2$ (Lecture 11, Theorem 10(c)); $\|u\|^2 = u'u$ is 0 only for $u = 0$ (Lecture 6, Definition 1); $u = Mc$:

$$X'c = a, \quad a'\hat{\beta} = c'Hy \text{ is linear unbiased} \qquad \text{(Theorem 2(a),(b))}$$
$$\operatorname{var}(c'y) = \sigma^2 c'c, \quad \operatorname{var}(c'Hy) = \sigma^2 c'HH'c = \sigma^2 c'Hc = \sigma^2 a'Ga$$
$$\operatorname{var}(c'y) - \operatorname{var}(a'\hat{\beta}) = \sigma^2 c'(I_n - H)c = \sigma^2 c'M'Mc = \sigma^2\|u\|^2 \geqslant 0$$
$$\sigma^2\|u\|^2 = 0 \implies u = 0 \implies c = Hc \implies c'y = c'Hy = a'\hat{\beta} \text{ (all } y)$$

□

So $\operatorname{var}(c'y) = \operatorname{var}(a'\hat\beta) + \sigma^2\|Mc\|^2$. Under (A4) $\operatorname{rank} X = p$, every $a'\beta$ is estimable (Theorem 2(a)): Theorem 4 gives $\operatorname{var}(a'\tilde\beta) \geqslant \operatorname{var}(a'\hat\beta)$ for linear unbiased $\tilde\beta$ of β (Lecture 11, Theorem 13).

Example 5. In Example 3, with the g-inverse $G = \operatorname{diag}(0, \frac13, \frac12, 1)$ of $X'X$ (Lecture 11, Example 11), under (A1), (A3): find $\beta^0 = GX'y$, $\mathsf{E}(\beta^0)$, $\operatorname{var}(\beta^0)$ and $\operatorname{var}(\bar y_1 - \bar y_2)$.

Solution.
$GX'X$: the rows of $X'X$ times $0, \frac13, \frac12, 1$ (Lecture 1, Proposition 13(a)); Lecture 9, Theorem 3(b):

$$GX'X = \begin{pmatrix} 0&0&0&0\\ 1&1&0&0\\ 1&0&1&0\\ 1&0&0&1 \end{pmatrix}, \qquad \beta^0 = GX'y = (0, \bar y_1, \bar y_2, \bar y_3)'$$

$$\begin{aligned} \mathsf{E}(\beta^0) &= GX'X\beta = (0,\ \mu+\alpha_1,\ \mu+\alpha_2,\ \mu+\alpha_3)' && ((\text{A1}))\\ \operatorname{var}(\beta^0) &= GX'(\sigma^2 I_n)XG' = \sigma^2 GX'XG = \sigma^2 \operatorname{diag}(0, \tfrac13, \tfrac12, 1) && ((\text{A3});\ G' = G)\\ \operatorname{var}(\bar y_1 - \bar y_2) &= a' \operatorname{var}(\beta^0) a = \sigma^2\left(\tfrac13 + \tfrac12\right) = \tfrac56\sigma^2 = \sigma^2 a'Ga && (a = (0,1,-1,0)') \end{aligned}$$

So β^0 is the $\hat\beta$ of Example 3, and $\bar y_l$ estimates $\mu + \alpha_l$, not α_l; $GX'XG$ has the columns of $GX'X$ times $0, \frac13, \frac12, 1$ (Lecture 1, Proposition 13(a)); and $\frac56\sigma^2$ is the variance of Theorem 4.

2. The normal linear model; t intervals and tests

Definition 6. The **normal linear model** is the linear model of Lecture 11, Definition 1, with $X \in \mathbb{R}^{n\times p}$ of any rank, under (A5) $\varepsilon \sim N_n(0, \sigma^2 I_n)$ for a number $\sigma^2 > 0$ (Lecture 10, Definition 1: multivariate normal). Then (A2), (A3) hold (Lecture 10, Proposition 2: $z \sim N_n(\mu, \Sigma) \Rightarrow \mathsf{E}(z) = \mu$, $\operatorname{var}(z) = \Sigma$), and $y = X\beta + I_n\varepsilon \sim N_n(X\beta, \sigma^2 I_n)$ (Lecture 10, Theorem 3: $Az+b \sim N_m(A\mu+b, A\Sigma A')$), so (A1) holds; and $y_1, \dots, y_n$ are independent (F1), $y_i \sim N(x_i'\beta, \sigma^2)$, x_i' the row i of X (Lecture 10, Theorem 6(b): $z \sim N_n(\mu, \operatorname{diag}(\sigma_1^2, \dots, \sigma_n^2)) \iff$ the z_i are independent $N(\mu_i, \sigma_i^2)$). Let $\alpha \in (0,1)$. A **hypothesis** H_0 is a condition on the parameters, such as $K\beta = m$. A **test** of H_0 is given by a **rejection region** $\mathcal{R}$, a set of possible values of y: H_0 is rejected if the observed y lies in $\mathcal{R}$. The test has **level** α if $\mathsf{P}(y \in \mathcal{R}) = \alpha$ for all parameter values that satisfy H_0. Usually $\mathcal{R}$ is described by a **test statistic** T computed from y, as $\{T > c\}$ or $\{|T| > c\}$. When H_0 is rejected at level α, the departure from H_0 is **significant** at level α. A **confidence interval** of level $1-\alpha$ for $\theta \in \mathbb{R}$ is an interval $[L, U]$ computed from y with $\mathsf{P}(L \leqslant \theta \leqslant U) = 1-\alpha$ for all parameter values (Lecture 10, Definition 18). The distributions χ^2_r, t_ν, $F_{a,b}$ and their quantiles $t_{\nu,\tau}$, $F_{a,b,\tau}$ are those of F2, F3.

Theorem 7. In the normal linear model let $\operatorname{rank} X = r < n$, G a g-inverse of $X'X$, $\beta^0 = GX'y$, $H = XGX'$, $M = I_n - H$, and let $K \in \mathbb{R}^{q\times p}$ be such that every entry of $K\beta$ is estimable.

(a) $K\beta^0 \sim N_q(K\beta, \sigma^2 KGK')$, and $K\beta^0$, KGK' are the same for every g-inverse G of $X'X$. Under (A4), $\hat\beta = (X'X)^{-1}X'y \sim N_p(\beta, \sigma^2(X'X)^{-1})$.

(b) $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-r}$, and $\mathsf{P}(\mathrm{SSE} = 0) = 0$.

(c) $K\beta^0$ and SSE are independent.

Proof.

(a) Row i of K is $t_i'X$, $t_i \in \mathbb{R}^n$ (Theorem 2(a)); $T \in \mathbb{R}^{q\times n}$ has the rows t_i', so $K = TX$ (Lecture 1, Theorem 4(c)); $y \sim N_n(X\beta, \sigma^2 I_n)$ (Definition 6):

$$\begin{aligned} K\beta^0 = TXGX'y = THy \sim N_q(THX\beta,\ \sigma^2 THH'T') && \text{(Lecture 10, Theorem 3)}\\ THX\beta = TX\beta = K\beta, \qquad THH'T' = THT' = TXGX'T' = KGK' && \text{(Lecture 11, Theorem 10(c))} \end{aligned}$$

H does not depend on G (Lecture 11, Theorem 10(b)), so neither do $K\beta^0 = THy$ and $KGK' = THT'$. Under (A4), $X'X$ is invertible and $(X'X)^{-1}$ is its only g-inverse (Lecture 11, Theorem 2(b), Definition 8), every $a'\beta$ is estimable (Theorem 2(a)), and $K = I_p$ gives $\hat{\beta}$.

(b) $y \sim N_n(X\beta, \sigma^2 I_n)$; Lecture 11, Theorem 10(c)–(e); $z \sim N_n(\mu, \sigma^2 I_n)$, $A' = A = A^2$, $\operatorname{rank} A = d \geqslant 1$, $A\mu = 0 \Rightarrow z'Az/\sigma^2 \sim \chi^2_d$ (Lecture 10, Theorem 15):

$$\begin{aligned} \mathrm{SSE} = y'My, \quad M' = M = M^2 & \\ \operatorname{rank} M = n - r \geqslant 1, \quad MX\beta = 0 & \\ \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-r} & \quad \text{(Lecture 10, Theorem 15, } z = y,\ A = M\text{)} \\ \mathsf{P}(\mathrm{SSE} = 0) = \mathsf{P}(\mathrm{SSE}/\sigma^2 = 0) = 0 & \quad \text{(F3)} \end{aligned}$$

(c) $K\beta^0 = (TH)y$ ((a)); for $z \sim N_n(\mu, \sigma^2 I_n)$, $A' = A = A^2$ and $KA = 0$, Kz and $z'Az$ are independent (Lecture 10, Theorem 16(b)); here $z = y$, and TH, M for K, A:

$$(TH)M = T(HM) = 0 \implies K\beta^0 \text{ and } \mathrm{SSE} = y'My \text{ independent} \qquad \text{(Lecture 11, Theorem 10(c))}$$

□

We write $s^2 = \hat{\sigma}^2 = \mathrm{SSE}/(n-r)$, unbiased for σ^2 (Lecture 11, Theorem 16: $\mathsf{E}(\mathrm{SSE}) = (n-r)\sigma^2$), and $s = \sqrt{s^2}$; by (b), $s > 0$ with probability 1.

Theorem 8. In the normal linear model let $\operatorname{rank} X = r < n$, G a g-inverse of $X'X$, $\beta^0 = GX'y$, and let $a \in \mathbb{R}^p$ be nonzero with $a'\beta$ estimable. Then $a'Ga > 0$; let $\mathrm{SE}(a'\beta^0) = s\sqrt{a'Ga}$, the **standard error** of $a'\beta^0$, and $t^* = t_{n-r,1-\alpha/2}$.

(a) $T = (a'\beta^0 - a'\beta)/\mathrm{SE}(a'\beta^0) \sim t_{n-r}$.

(b) $\left[a'\beta^0 - t^*\,\mathrm{SE}(a'\beta^0),\ a'\beta^0 + t^*\,\mathrm{SE}(a'\beta^0)\right]$ is a confidence interval of level $1-\alpha$ for $a'\beta$. For $m_0 \in \mathbb{R}$, the test of $H_0\colon a'\beta = m_0$ that rejects when $|T_0| > t^*$, where $T_0 = (a'\beta^0 - m_0)/\mathrm{SE}(a'\beta^0)$, has level α; it rejects exactly when m_0 lies outside this interval.

$\mathrm{SE}(a'\beta^0)$ and the interval are the same for every g-inverse G of $X'X$ (Theorem 7(a), $K = a'$; s by Lecture 11, Theorem 10(e)). Under (A4), $G = (X'X)^{-1}$, $\beta^0 = \hat{\beta}$ and $r = p$.

Proof.

$a'Ga > 0$: $a = X'c$ (Theorem 2(a)); $H = XGX'$:

$$\begin{aligned} a'Ga = c'XGX'c = c'Hc = c'H'Hc = \|Hc\|^2 & \quad \text{(Lecture 11, Theorem 10(c))} \\ X'Hc = X'c = a \neq 0 \implies Hc \neq 0 \implies a'Ga > 0 & \quad \text{(Lecture 6, Definition 1)} \end{aligned}$$

(a) Lecture 10, Theorem 3 for z; $s > 0$ with probability 1:

$$\begin{aligned} z = \frac{a'\beta^0 - a'\beta}{\sigma\sqrt{a'Ga}} \sim N(0,1) & \quad \text{(Theorem 7(a), } K = a'\text{)} \\ v = \mathrm{SSE}/\sigma^2 \sim \chi^2_{n-r}; \quad z,\ v \text{ independent} & \quad \text{(Theorem 7(b),(c); F1)} \\ T = \frac{z}{s/\sigma} = \frac{z}{\sqrt{v/(n-r)}} \sim t_{n-r} & \quad \text{(F2)} \end{aligned}$$

(b) For $T \sim t_\nu$: $\mathsf{P}(-t_{\nu,1-\alpha/2} \leqslant T \leqslant t_{\nu,1-\alpha/2}) = 1-\alpha$ (Lecture 10, Theorem 11(a)) and $t_{\nu,1-\alpha/2} > 0$ (Lecture 10, Theorem 11(b)); $\mathrm{SE} = \mathrm{SE}(a'\beta^0) > 0$ with probability 1:

$$\begin{aligned} \mathsf{P}(|T| \leqslant t^*) = 1-\alpha \quad (T \sim t_{n-r} \text{ by (a)}) & \quad \text{(Lecture 10, Theorem 11(a))} \\ |T| \leqslant t^* \iff a'\beta^0 - t^*\,\mathrm{SE} \leqslant a'\beta \leqslant a'\beta^0 + t^*\,\mathrm{SE} & \quad \text{(Definition 6)} \\ H_0\colon a'\beta = m_0 \implies T_0 = T & \\ \mathsf{P}(|T_0| > t^*) = 1 - (1-\alpha) = \alpha \text{ under } H_0 & \quad \text{(F1)} \\ |T_0| > t^* \iff |a'\beta^0 - m_0| > t^*\,\mathrm{SE} \iff m_0 \text{ outside the interval} & \end{aligned}$$

□

Example 9. In Example 3, assume the normal linear model. Find the confidence interval of level $1-\alpha$ for $\alpha_1-\alpha_2$, and the test of $H_0\colon \alpha_1=\alpha_2$ at level α.

Solution.

$\alpha_1-\alpha_2$ is estimable (Example 3); β^0, $a'Ga$ as in Example 5; SSE $=4$ (Lecture 11, Example 11), $r=\operatorname{rank}X=3$ (Lecture 11, Theorem 6(b)); Theorem 8(b):

$$a=(0,1,-1,0)'\colon\quad a'\beta^0=6-4=2,\quad a'Ga=\tfrac13+\tfrac12=\tfrac56;\quad \text{SSE}=4,\ n-r=6-3=3,\ s^2=\tfrac43$$
$$\text{SE}=s\sqrt{5/6}=\sqrt{10/9}=\sqrt{10}/3;\quad \text{interval } \left[2-t_{3,1-\alpha/2}\sqrt{10}/3,\ 2+t_{3,1-\alpha/2}\sqrt{10}/3\right]$$
$$m_0=0\colon\quad T_0=2/(\sqrt{10}/3)=6/\sqrt{10};\quad \text{the test rejects } H_0 \iff t_{3,1-\alpha/2}<6/\sqrt{10}$$

Equivalently, the interval contains 0 exactly when $t_{3,1-\alpha/2}\geqslant 6/\sqrt{10}$.

3. Testable hypotheses and the F test

Definition 10. Let $\operatorname{rank}X=r$, $K\in\mathbb{R}^{q\times p}$ and $m\in\mathbb{R}^q$. The hypothesis $H_0\colon K\beta=m$ is **testable** if every entry of $K\beta$ is estimable and $\operatorname{rank}K=q$ (the rows of K are linearly independent).

Let $H_0\colon K\beta=m$ be testable, G a g-inverse of $X'X$, $H=XGX'$; $K=T_1X$, $T_1\in\mathbb{R}^{q\times n}$ (Theorem 2(a); Lecture 1, Theorem 4(c)); for $A\in\mathbb{R}^{n\times p}$, $A'A$ is PD (Lecture 8, Definition 13: symmetric, $x'Ax>0$ for $x\neq 0$) $\iff$ $\operatorname{rank}A=p$ (Lecture 8, Theorem 15(b)), and a PD matrix is invertible (Lecture 8, Theorem 14(c)):

$$T=T_1H\colon\quad TX=T_1HX=T_1X=K,\quad TH=T_1H^2=T \qquad \text{(Lecture 11, Theorem 10(c))}$$
$$q=\operatorname{rank}(TX)\leqslant\operatorname{rank}T\leqslant q \implies \operatorname{rank}T'=\operatorname{rank}T=q \qquad \text{(Lecture 4, Theorems 15(a), 12(a))}$$
$$V=KGK'=TXGX'T'=THT'=THH'T'=TT' \qquad \text{(Lecture 11, Theorem 10(c))}$$
$$\implies V \text{ is PD } (A=T'), \text{ hence invertible} \qquad \text{(Lecture 8, Theorems 15(b), 14(c))}$$

$T=T_1H$ and $V=THT'$ do not depend on G (Lecture 11, Theorem 10(b)). Under (A4), $H_0\colon K\beta=m$ is testable for every K of rank q (Theorem 2(a)).

Theorem 11. In the normal linear model let $\operatorname{rank}X=r<n$, G a g-inverse of $X'X$, $\beta^0=GX'y$, and $H_0\colon K\beta=m$ testable, with $K\in\mathbb{R}^{q\times p}$ and $V=KGK'$.

(a) Let $F=\left[(K\beta^0-m)'V^{-1}(K\beta^0-m)/q\right]/s^2$, defined with probability 1 (V invertible by the lines after Definition 10, $s^2>0$ by Theorem 7(b)) and the same for every g-inverse G of $X'X$. If H_0 holds, then $F\sim F_{q,n-r}$; so the test that rejects H_0 when $F>F_{q,n-r,1-\alpha}$ has level α.

(b) If $q=1$ and $K=a'$, then $F=T_0^2$, with T_0 of Theorem 8(b) for $m_0=m$; the test of (a) rejects exactly when that of Theorem 8(b) does.

Proof. (a) $K\beta^0$ and V do not depend on G (Theorem 7(a), the lines after Definition 10), nor does $s^2=y'My/(n-r)$ (Lecture 11, Theorem 10(b),(e)).

Algebra: T as after Definition 10 ($TH=T$, $V=TT'$); H_0 holds; $\varepsilon=y-X\beta$; $MX=0$, $X'M=0$ (Lecture 11, Theorem 10(c)); $A'=A=A^2\Rightarrow\operatorname{rank}A=\operatorname{tr}A$ (Lecture 8, Theorem 11):

$$K\beta^0=TXGX'y=THy=Ty$$
$$K\beta^0-m=Ty-TX\beta=T\varepsilon \qquad (m=K\beta=TX\beta)$$
$$U=T'V^{-1}T=U',\quad U^2=T'V^{-1}(TT')V^{-1}T=U \qquad \text{(Lecture 2, Corollary 12)}$$
$$\operatorname{rank}U=\operatorname{tr}U=\operatorname{tr}(V^{-1}TT')=\operatorname{tr}I_q=q \qquad \text{(Lecture 1, Theorem 10(b))}$$
$$TM=T-TH=0 \implies UM=0$$
$$Q=(K\beta^0-m)'V^{-1}(K\beta^0-m)=\varepsilon'U\varepsilon;\ \text{SSE}=y'My=\varepsilon'M\varepsilon$$

Distribution: $\varepsilon \sim N_n(0, \sigma^2 I_n)$ (A5); $M' = M = M^2$, $\operatorname{rank} M = n - r$ (Lecture 11, Theorem 10(c),(d)); for $z \sim N_n(\mu, \sigma^2 I_n)$ and symmetric A, B with $A^2 = A$, $B^2 = B$, $AB = 0$, $z'Az$ and $z'Bz$ are independent (Lecture 10, Theorem 16(a)):

$$\begin{aligned} Q/\sigma^2 = \varepsilon' U \varepsilon/\sigma^2 \sim \chi^2_q & \qquad \text{(Lecture 10, Theorem 15, } \mu = 0) \\ \mathrm{SSE}/\sigma^2 = \varepsilon' M \varepsilon/\sigma^2 \sim \chi^2_{n-r} & \qquad \text{(Lecture 10, Theorem 15, } \mu = 0) \\ Q \text{ and SSE independent} & \qquad \text{(Lecture 10, Theorem 16(a), } UM = 0) \\ F = \frac{(Q/\sigma^2)/q}{(\mathrm{SSE}/\sigma^2)/(n-r)} \sim F_{q,n-r} & \qquad \text{(F1; F2)} \\ \mathsf{P}(F > F_{q,n-r,1-\alpha}) = 1 - (1-\alpha) = \alpha & \qquad \text{(F3, F1)} \end{aligned}$$

(b) $K = a'$, $V = a'Ga$, $\mathrm{SE}(a'\beta^0)^2 = s^2 a'Ga$ (Theorem 8):

$$F = \frac{(a'\beta^0 - m)^2}{s^2 a'Ga} = \left(\frac{a'\beta^0 - m}{\mathrm{SE}(a'\beta^0)}\right)^2 = T_0^2$$

and $F > F_{1,n-r,1-\alpha} \iff T_0^2 > (t^*)^2 \iff |T_0| > t^*$ (Lecture 10, Theorem 11(b): $t^2_{\nu,1-\alpha/2} = F_{1,\nu,1-\alpha}$, $t_{\nu,1-\alpha/2} > 0$). □

Example 12. Eight plots receive a fertilizer at two levels, coded $x_{i1} = -1, 1$, and irrigation at two levels, coded $x_{i2} = -1, 1$, two plots for each combination; the yields are

x_{i1}	−1	−1	−1	−1	1	1	1	1
x_{i2}	−1	−1	1	1	−1	−1	1	1
y_i	9	3	9	7	11	13	11	17

In the normal linear model $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$ (multiple regression with $n = 8$, $k = 2$, Lecture 11, Definition 1), find the confidence interval of level $1 - \alpha$ for β_1, and the tests of $\beta_2 = 0$ and of $\beta_1 = \beta_2 = 0$ at level α. Put $t^* = t_{5,1-\alpha/2}$.

Solution.

The three columns of X are orthogonal with squared length 8 (Lecture 1, Theorem 4(d)); $X'X$ invertible $\iff$ $\operatorname{rank} X = 3$ (Lecture 11, Theorem 2(b)); under (A4), $G = (X'X)^{-1}$, $\beta^0 = \hat\beta$, $r = 3$ (Theorem 8); every $a'\beta$ is estimable (Theorem 2(a)):

$$\begin{gathered} X'X = 8I_3, \quad \operatorname{rank} X = 3, \quad G = (X'X)^{-1} = \tfrac{1}{8} I_3, \quad X'y = (80, 24, 8)', \quad \hat\beta = \tfrac{1}{8} X'y = (10, 3, 1)' \\ \hat y_i = 10 + 3x_{i1} + x_{i2} = 6,\ 8,\ 12,\ 14 \text{ for the four combinations} \\ e = y - \hat y = (3, -3, 1, -1, -1, 1, -3, 3)', \quad \mathrm{SSE} = e'e = 40, \quad s^2 = 40/(8-3) = 8 \end{gathered}$$

$\mathrm{SE}(\hat\beta_j) = s\sqrt{1/8} = 1$ ($a'Ga = \frac{1}{8}$ for $a = \mathrm{col}_{j+1}(I_3)$); Theorems 8(b), 11(b):

$$\begin{gathered} \beta_1 : \text{ interval } [3 - t^*,\ 3 + t^*] \\ \beta_2 = 0 : \ T_0 = (1-0)/1 = 1; \text{ the test rejects } \iff t^* < 1; \quad F = T_0^2 = 1 \end{gathered}$$

$\beta_1 = \beta_2 = 0$ is $K\beta = 0$, $K = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right)$, $\operatorname{rank} K = 2$: testable under (A4) (the lines after Definition 10); Theorem 11(a), $q = 2$:

$$\begin{gathered} K\hat\beta = (3, 1)', \quad V = KGK' = \tfrac{1}{8} I_2, \quad V^{-1} = 8I_2, \quad (K\hat\beta)' V^{-1} K\hat\beta = 8(9 + 1) = 80 \\ F = \frac{80/2}{8} = 5 : \text{ the test rejects } H_0 \iff F_{2,5,1-\alpha} < 5 \end{gathered}$$

With $\bar y = 10$: $\mathrm{SSR} = 2(16 + 4 + 4 + 16) = 80$, $\mathrm{SST} = 120$ and $R^2 = \frac{2}{3}$ (Lecture 11, Definition 17: $\mathrm{SSR} = \sum_i (\hat y_i - \bar y)^2$, $\mathrm{SST} = \sum_i (y_i - \bar y)^2$, $R^2 = \mathrm{SSR}/\mathrm{SST}$); so here $(K\hat\beta)' V^{-1} K\hat\beta = 80 = \mathrm{SSR}$.

4. One-way analysis of variance

Theorem 13. In the one-way design (Lecture 11, Theorem 6: n observations in g groups of sizes $n_l \geqslant 1$, the indicator $d_l \in \mathbb{R}^n$ of group l, $X = [\mathbf{1}_n\ d_1\ \cdots\ d_g]$, $X_1 = [d_1\ \cdots\ d_g]$), let $\mathcal{I}_l$ be the set of the observations (row indices) of group l, $\bar{y}_l = n_l^{-1} \sum_{i \in \mathcal{I}_l} y_i$ the **group mean** and $\bar{y}$ the **grand mean** of all n observations. Then $\hat{y} = \bar{y}_1 d_1 + \cdots + \bar{y}_g d_g$ (the fitted value of each observation is the mean of its group), and

$$\sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{l=1}^{g} n_l (\bar{y}_l - \bar{y})^2 + \sum_{l=1}^{g} \sum_{i \in \mathcal{I}_l} (y_i - \bar{y}_l)^2$$

that is, SST = SSR + SSE, with the **between-group** sum of squares SSR and the **within-group** sum of squares SSE.

Proof.

Fitted values: X_1 has full column rank (Lecture 11, Theorem 6(a)); $X_1' y$ holds the group totals $d_l' y$ (Lecture 1, Theorem 4(d)); no observation lies in two groups; $\operatorname{diag}(n_l)^{-1} = \operatorname{diag}(1/n_l)$ (Lecture 1, Proposition 13(b); Lecture 2, Theorem 15):

$$\begin{aligned}
d_l' d_m = 0\ (l \neq m), \quad d_l' d_l = n_l \implies X_1' X_1 &= \operatorname{diag}(n_1, \ldots, n_g) && \text{(Lecture 1, Theorem 4(d))} \\
\hat{\beta}_1 = (X_1' X_1)^{-1} X_1' y &= (\bar{y}_1, \ldots, \bar{y}_g)' && \text{(Lecture 11, Theorem 3(c))} \\
X_1 \hat{\beta}_1 = \bar{y}_1 d_1 + \cdots + \bar{y}_g d_g &= \hat{y} \quad (\mathcal{C}(X_1) = \mathcal{C}(X)) && \text{(Lecture 11, Theorems 6(c), 3(b))}
\end{aligned}$$

Sums of squares: $\hat{y}_i = \bar{y}_l$ for $i \in \mathcal{I}_l$:

$$\begin{aligned}
\mathrm{SSR} &= \textstyle\sum_i (\hat{y}_i - \bar{y})^2 = \sum_l n_l (\bar{y}_l - \bar{y})^2 \\
\mathrm{SSE} &= \textstyle\sum_i (y_i - \hat{y}_i)^2 = \sum_l \sum_{i \in \mathcal{I}_l} (y_i - \bar{y}_l)^2 && \text{(Lecture 11, Definition 17)} \\
\mathrm{SST} &= \mathrm{SSR} + \mathrm{SSE} \quad (\mathbf{1}_n = \textstyle\sum_l d_l \in \mathcal{C}(X)) && \text{(Lecture 11, Theorem 18)}
\end{aligned}$$

□

Corollary 14. In the one-way design of Lecture 11, Theorem 6, use the design $X_1 = [d_1\ \cdots\ d_g]$ with $\beta = (\mu_1, \ldots, \mu_g)'$ (the group means), and assume the normal linear model with $n > g \geqslant 2$. With SSR, SSE of Theorem 13, let $F = \big[\mathrm{SSR}/(g-1)\big] / \big[\mathrm{SSE}/(n-g)\big]$, defined with probability 1 (Theorem 7(b)). If $H_0 \colon \mu_1 = \cdots = \mu_g$ holds, then $F \sim F_{g-1,n-g}$; so the test that rejects H_0 when $F > F_{g-1,n-g,1-\alpha}$ has level α.

Proof. $\bar{J}_n = \frac{1}{n}\mathbf{1}\mathbf{1}'$ (Lecture 1, Definition 14), $\bar{J}_n' = \bar{J}_n = \bar{J}_n^2$, $\operatorname{tr} \bar{J}_n = 1$ (Lecture 1, Proposition 15), $\bar{J}_n y = \bar{y}\mathbf{1}_n$ ($\bar{y} = \frac{1}{n}\mathbf{1}'y$, Lecture 1, Example 16); $\mathbf{1}_n = X_1 \mathbf{1}_g$.

$\operatorname{rank} X_1 = g$ (Lecture 11, Theorem 6(a)), so $H = X_1 (X_1' X_1)^{-1} X_1'$ and $M = I_n - H$ have $H' = H = H^2$, $HX_1 = X_1$, $HM = 0$, $\operatorname{tr} H = g$, $Hy = \hat{y}$ (Lecture 11, Theorems 2(b), 10, design X_1); $A = H - \bar{J}_n$:

$$\begin{aligned}
H\bar{J}_n = \tfrac{1}{n}(HX_1\mathbf{1}_g)\mathbf{1}' = \tfrac{1}{n}\mathbf{1}\mathbf{1}' &= \bar{J}_n && \text{(Lecture 1, Theorem 3(a))} \\
\bar{J}_n H = (H\bar{J}_n)' &= \bar{J}_n && \text{(Lecture 1, Theorem 8)} \\
A' = A, \quad A^2 = H - 2\bar{J}_n + \bar{J}_n^2 &= A \\
\operatorname{rank} A = \operatorname{tr} A = \operatorname{tr} H - \operatorname{tr} \bar{J}_n &= g - 1 && \text{(Lecture 8, Theorem 11; Lecture 1, Theorem 10(a))} \\
AM = HM - \bar{J}_n(I_n - H) &= -\bar{J}_n + \bar{J}_n = 0 \\
y'Ay = y'A'Ay &= \|\hat{y} - \bar{y}\mathbf{1}_n\|^2 = \mathrm{SSR} && \text{(Theorem 13)} \\
y'My = e'e &= \mathrm{SSE} && \text{(Lecture 11, Theorem 10(e))}
\end{aligned}$$

Distribution: $y \sim N_n(X_1\beta, \sigma^2 I_n)$ (Definition 6); $g-1 \geqslant 1$; $\bar{J}_n \mathbf{1}_n = \frac{1}{n}\mathbf{1}(\mathbf{1}'\mathbf{1}) = \mathbf{1}_n$ (Lecture 1, Proposition 15):

$$AX_1\beta = \mu_1 A(d_1 + \cdots + d_g) = \mu_1(H\mathbf{1}_n - \bar{J}_n\mathbf{1}_n) = \mu_1(\mathbf{1}_n - \mathbf{1}_n) = 0 \quad (H_0)$$
$$\mathrm{SSR}/\sigma^2 = y'Ay/\sigma^2 \sim \chi^2_{g-1} \quad \text{(Lecture 10, Theorem 15)}$$
$$\mathrm{SSE}/\sigma^2 = y'My/\sigma^2 \sim \chi^2_{n-g} \quad \text{(Theorem 7(b), } r = g\text{)}$$
$$\text{SSR and SSE are independent } (AM = 0) \quad \text{(Lecture 10, Theorem 16(a))}$$
$$F = \frac{(\mathrm{SSR}/\sigma^2)/(g-1)}{(\mathrm{SSE}/\sigma^2)/(n-g)} \sim F_{g-1,n-g},\ \mathsf{P}(F > F_{g-1,n-g,1-\alpha}) = \alpha \quad \text{(F1–F3)}$$

□

The computations of Corollary 14 are arranged in the **analysis-of-variance table**: the degrees of freedom are those of the χ^2 distributions in the proof, and $n-1 = (g-1)+(n-g)$ for SST; the **mean squares** are the sums of squares divided by them, and F is their ratio:

source	degrees of freedom	sum of squares	mean square
between groups	$g-1$	$\mathrm{SSR} = \sum_l n_l(\bar{y}_l - \bar{y})^2$	$\mathrm{SSR}/(g-1)$
within groups	$n-g$	$\mathrm{SSE} = \sum_l \sum_{i\in\mathcal{I}_l}(y_i - \bar{y}_l)^2$	$s^2 = \mathrm{SSE}/(n-g)$
total	$n-1$	$\mathrm{SST} = \sum_i (y_i - \bar{y})^2$	

Example 15. In Example 3, assume the normal linear model. Test H_0: $\alpha_1 = \alpha_2 = \alpha_3$ at level α.

Solution.

Group means $6, 4, 2$ (Lecture 11, Example 7); $e = (-1, 1, 0, -1, 1, 0)'$, $\mathrm{SSE} = 4$ (Lecture 11, Example 11); Theorem 13:

$$\bar{y} = 28/6 = 14/3; \quad \mathrm{SSE} = 4,\ r = \operatorname{rank} X = 3,\ s^2 = 4/3$$
$$\mathrm{SSR} = 3\Big(6 - \frac{14}{3}\Big)^2 + 2\Big(4 - \frac{14}{3}\Big)^2 + \Big(2 - \frac{14}{3}\Big)^2 = \frac{48+8+64}{9} = \frac{40}{3}, \qquad \mathrm{SST} = \frac{40}{3} + 4 = \frac{52}{3}$$

$\mathbf{1}_6 = d_1 + d_2 + d_3$; Corollary 14 with $n = 6 > g = 3$:

$$X\beta = \mu\mathbf{1}_6 + \alpha_1 d_1 + \alpha_2 d_2 + \alpha_3 d_3 = X_1(\mu + \alpha_1,\ \mu + \alpha_2,\ \mu + \alpha_3)' :\ H_0 \text{ is } \mu_1 = \mu_2 = \mu_3$$
$$F = \frac{(40/3)/2}{4/3} = 5, \text{ with 2 and 3 degrees of freedom: the test rejects } H_0 \iff F_{2,3,1-\alpha} < 5$$

source	degrees of freedom	sum of squares	mean square
between groups	2	40/3	20/3
within groups	3	4	4/3
total	5	52/3	

The same F by Theorem 11(a) in the design X ($r = 3$): H_0 is $K\beta = 0$, testable (the criterion of Example 3 for the rows); G, $\beta^0 = (0, 6, 4, 2)'$ as in Example 5; the rows of K are independent $\iff$ $\operatorname{rank} K = 2$ (Lecture 4, Theorem 12(c)); V^{-1} by Lecture 2, Proposition 17:

$$K = \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix} :\ -0+1-1+0 = 0,\ -0+0+1-1 = 0 :\ \text{rows estimable}$$
$$c_1(0,1,-1,0) + c_2(0,0,1,-1) = (0,\ c_1,\ c_2 - c_1,\ -c_2) = 0 \implies c_1 = c_2 = 0 :\ \operatorname{rank} K = 2$$
$$K\beta^0 = (2,2)', \quad V = KGK' = \begin{pmatrix} \frac{5}{6} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix}$$
$$ad - bc = \tfrac{5}{4} - \tfrac{1}{4} = 1, \quad V^{-1} = \begin{pmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{5}{6} \end{pmatrix}$$
$$(K\beta^0)'V^{-1}K\beta^0 = 4\Big(\tfrac{3}{2} + \tfrac{1}{2} + \tfrac{1}{2} + \tfrac{5}{6}\Big) = \tfrac{40}{3} = \mathrm{SSR}, \qquad F = \frac{40/3}{2 \cdot 4/3} = 5$$

Uncentered, $y'y = n\bar{y}^2 + \text{SSR} + \text{SSE}$ (Lecture 11, Theorem 18): $n\bar{y}^2 = 6 \cdot \frac{196}{9} = \frac{392}{3}$, and $y'y = 148 = \frac{392}{3} + \frac{40}{3} + 4$.

Practice. The Exercises and Problems of the textbook, Chapter 14: estimable functions and their best linear unbiased estimators, the normal linear model, t intervals and tests, testable hypotheses and the F test, and the one-way analysis of variance.